POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

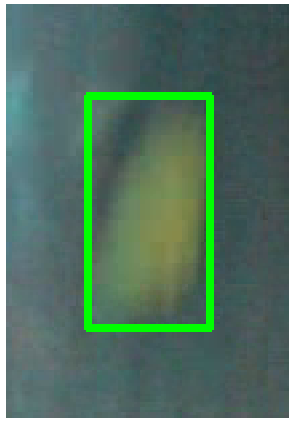
Video ID:	26c197fc-9b0a-47da-a44c-4cff4ed3b95d
Total Frames:	11951
Frame Rate:	25.0 fps (assumed)
Duration:	478.0 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	4257	✓
RT-DETR Detections	7098	✓
MedSAM2 Detections	7590	✓
Consensus (3-Model Agreement)	5	✓
Risk Classification	**Count**	**Percentage**
HIGH RISK	1	20.0%
MEDIUM RISK	2	40.0%
LOW RISK	2	40.0%
TOTAL ANALYZED	5	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 26 frames)

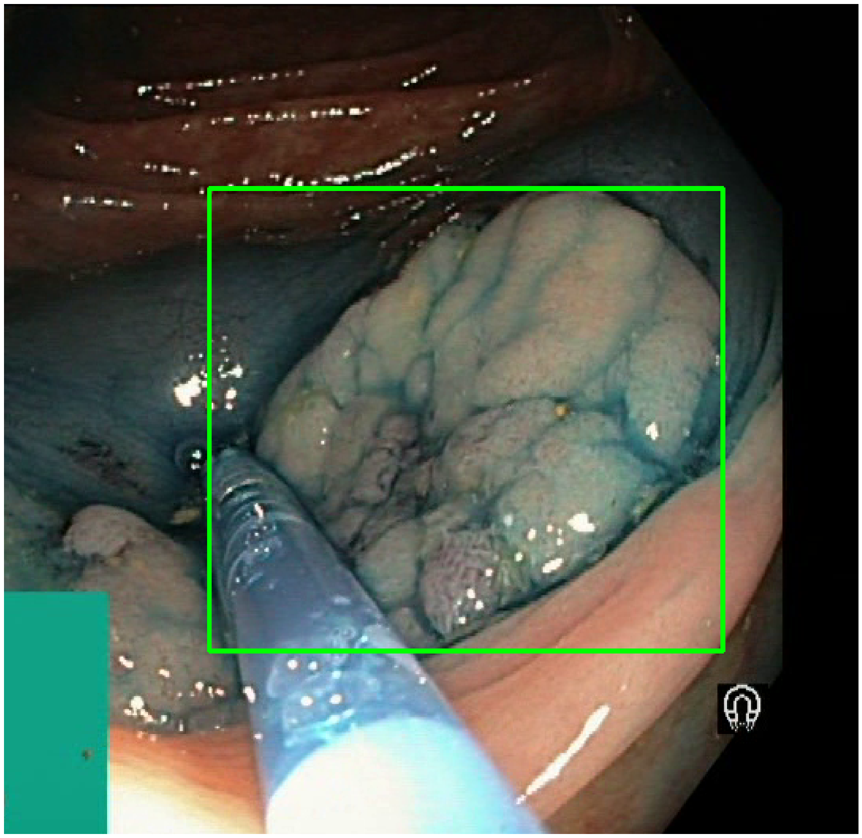


Feature	Value	Interpretation
Frame Index	7515	Frame 7515 of 11951
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	MedSAM2 Unavailable
POLYP TYPE	**FLAT_POLYP**	Type confidence: 72.1%
Redness Score	0.081	Color-based vascularization
Relative Radius	20.7%	Polyp size as % of frame diagonal
Texture Score	0.047	Surface morphology
Vessel Visibility	0.190	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	83.4%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #2 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 2096 frames)



Feature	Value	Interpretation
Frame Index	6286	Frame 6286 of 11951
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**FLAT_POLYP**	Type confidence: 88.3%
Redness Score	0.154	Color-based vascularization
Relative Radius	25.7%	Polyp size as % of frame diagonal
Texture Score	0.191	Surface morphology
Vessel Visibility	0.253	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified

Detection Confidence	83.0%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #3 — LIFTED_POLYP [LOW_RISK] (Tracked across 327 frames)

Feature	Value	Interpretation
Frame Index	6400	Frame 6400 of 11951
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 69.3%
Redness Score	0.001	Color-based vascularization
Relative Radius	16.3%	Polyp size as % of frame diagonal
Texture Score	0.371	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	81.9%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #4 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 155 frames)

Feature	Value	Interpretation
Frame Index	11617	Frame 11617 of 11951
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**BLEEDING_POLYP**	Type confidence: 92.0%
Redness Score	0.209	Color-based vascularization
Relative Radius	12.6%	Polyp size as % of frame diagonal
Texture Score	0.428	Surface morphology
Vessel Visibility	0.755	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	MALIGNANT_POLYP	High suspicion for malignancy — biopsy and urgent resection recommended
Detection Confidence	79.1%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #5 — LIFTED_POLYP [LOW_RISK] (Tracked across 216 frames)

Feature	Value	Interpretation
Frame Index	3193	Frame 3193 of 11951
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	MedSAM2 Unavailable
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 74.5%
Redness Score	0.051	Color-based vascularization
Relative Radius	15.0%	Polyp size as % of frame diagonal
Texture Score	0.381	Surface morphology
Vessel Visibility	0.011	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified

Detection Confidence	88.1%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

CLINICAL IMPRESSION

Summary:
A total of 5 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:
• HIGH RISK polyps: 1 (20.0%)
• MEDIUM RISK polyps: 2 (40.0%)
• LOW RISK polyps: 2 (40.0%)

Recommendation:
The presence of HIGH RISK polyps warrants close clinical attention. Consider endoscopic resection or follow-up surveillance depending on clinical context.

Technical Details:
• Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
• Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
• Classification: Cluster-specific Decision Tree experts
• Confidence: Based on deep learning + medical heuristics

Disclaimer: *This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.*

AI-Based Polyp Detection System
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